

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics

#### Chapter 1: Set Language

#### 2 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                        |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Write the set of letters of the following words in Roster form: (i) ASSESSMENT<br>(ii) PRINCIPAL                                                                                                                           |
| 2     | Which of the following are sets? (i) The collection of prime numbers upto 100.<br>(ii) The collection of rich people in India. (iii) The collection of all rivers in India.<br>(iv) The collection of good Hockey players. |
| 3     | List the set of letters of the following words in Roster form: (i) INDIA (ii) PARALLELOGRAM<br>(iii) MISSISSIPPI (iv) CZECHOSLOVAKIA                                                                                       |
| 4     | Represent the following sets in Roster form:<br>(i) A = The set of all even natural numbers less than 20.<br>(ii) $B = \{y : y = \frac{1}{2n}, n \in \mathbb{N}, n \leq 5\}$                                               |
| 5     | Represent the following sets in Roster form: (i) $C = \{x : x \text{ is perfect cube, } 27 < x < 216\}$<br>(ii) $D = \{x : x \in \mathbb{Z}, -5 < x \leq 2\}$                                                              |
| 6     | Represent the following sets in set builder form:<br>(i) B = The set of all Cricket players in India who scored double centuries in One day Internationals.<br>(ii) $C = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$ |
| 7     | Represent the following sets in set builder form:<br>(i) D = The set of all tamil months in a year.<br>(ii) E = The set of odd Whole numbers less than 9.                                                                  |
| 8     | Represent the following sets in descriptive form: (i) $P = \{\text{January, June, July}\}$ (ii) $Q = \{7, 11, 13, 17, 19, 23, 29\}$                                                                                        |
| 9     | Represent the following sets in descriptive form: (i) $R = \{x : x \in \mathbb{N}, x < 5\}$<br>(ii) $S = \{x : x \text{ is a consonant in English alphabet}\}$                                                             |
| 10    | If $A = \{1, 2, 3, 4, 5, 7, 9, 11\}$ , find $n(A)$ .                                                                                                                                                                       |
| 11    | Are $P = \{x : -3 \leq x \leq 0, x \in \mathbb{Z}\}$ and $Q = \text{The set of all prime factors of 210, equivalent sets?}$                                                                                                |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12    | Are $A = \{x : x \in \mathbb{N}, 4 \leq x \leq 8\}$ and $B = \{4, 5, 6, 7, 8\}$ equal sets?                                                                                                                                 |
| 13    | Write all the subsets of $A = \{a, b\}$ .                                                                                                                                                                                   |
| 14    | Verify whether $A = \{20, 22, 23, 24\}$ and $B = \{25, 30, 40, 45\}$ are disjoint sets.                                                                                                                                     |
| 15    | Find the number of subsets and the number of proper subsets of a set $X = \{a, b, c, x, y, z\}$ .                                                                                                                           |
| 16    | Find the cardinal number of the following sets: (i) $M = \{p, q, r, s, t, u\}$ (ii) $P = \{x : x = 3n + 2, n \in \mathbb{W} \text{ and } x < 15\}$                                                                          |
| 17    | Find the cardinal number of the following sets: (i) $Q = \{y : y = \frac{4}{3n}, n \in \mathbb{N} \text{ and } 2 < n \leq 5\}$ (ii) $R = \{x : x \text{ is an integers, } x \in \mathbb{Z} \text{ and } -5 \leq x < 5\}$    |
| 18    | Identify the following sets as finite or infinite: (i) $X =$ The set of all districts in Tamilnadu.<br>(ii) $Y =$ The set of all straight lines passing through a point.                                                    |
| 19    | Identify the following sets as finite or infinite: (i) $A = \{x : x \in \mathbb{Z} \text{ and } x < 5\}$ (ii) $B = \{x : x^2 - 5x + 6 = 0, x \in \mathbb{N}\}$                                                              |
| 20    | Which of the following sets are equivalent or unequal or equal sets? $A =$ The set of vowels in the English alphabets, $B =$ The set of all letters in the word "VOWEL".                                                    |
| 21    | Which of the following sets are equivalent or unequal or equal sets? $C = \{2, 3, 4, 5\}$ , $D = \{x : x \in \mathbb{W}, 1 < x < 5\}$ .                                                                                     |
| 22    | Identify the following sets as null set or singleton set: (i) $A = \{x : x \in \mathbb{N}, 1 < x < 2\}$ (ii) $B =$ The set of all even natural numbers which are not divisible by 2.                                        |
| 23    | State which pairs of sets are disjoint or overlapping: (i) $A = \{f, i, a, s\}$ and $B = \{a, n, f, h, s\}$<br>(ii) $C = \{x : x \text{ is a prime number, } x > 2\}$ and $D = \{x : x \text{ is an even prime number}\}$ . |
| 24    | If $A = \{a, \{a, b\}\}$ , write all the subsets of $A$ .                                                                                                                                                                   |
| 25    | Write down the power set of the following sets: (i) $A = \{a, b\}$ (ii) $B = \{1, 2, 3\}$                                                                                                                                   |
| 26    | Write down the power set of the following sets: (i) $D = \{p, q, r, s\}$ (ii) $E = \emptyset$                                                                                                                               |
| 27    | Find the number of subsets and the number of proper subsets of the following sets: (i) $W = \{\text{red, blue, yellow}\}$ (ii) $X = \{x^2 : x \in \mathbb{N}, x^2 \leq 100\}$                                               |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28    | (i) If $n(A) = 4$ , find $n(P(A))$ . (ii) If $n(A) = 0$ , find $n(P(A))$ . (iii) If $n(P(A)) = 256$ , find $n(A)$ .                                                                                 |
| 29    | If $U = \{c, d, e, f, g, h, i, j\}$ and $A = \{c, d, g, j\}$ , find $A'$ .                                                                                                                          |
| 30    | If $P = \{m, n\}$ and $Q = \{m, i, j\}$ , then, represent $P$ and $Q$ in Venn diagram and hence find $P \cup Q$ .                                                                                   |
| 31    | Let $A = \{x : x \text{ is an even natural number and } 1 < x \leq 12\}$ and $B = \{x : x \text{ is a multiple of 3, } x \in \mathbb{N} \text{ and } x \leq 12\}$ . Find $A \cap B$ .               |
| 32    | If $A = \{2, 3\}$ and $C = \emptyset$ , find $A \cap C$ .                                                                                                                                           |
| 33    | If $A = \{-3, -2, 1, 4\}$ and $B = \{0, 1, 2, 4\}$ , find (i) $A - B$ (ii) $B - A$ .                                                                                                                |
| 34    | If $A = \{6, 7, 8, 9\}$ and $B = \{8, 10, 12\}$ , find $A \Delta B$ .                                                                                                                               |
| 35    | Find $A \cup B, A \cap B, A - B$ and $B - A$ for the sets: $A = \{2, 6, 10, 14\}$ and $B = \{2, 5, 14, 16\}$                                                                                        |
| 36    | Find $A \cup B, A \cap B, A - B$ and $B - A$ for the sets: $A = \{a, b, c, e, u\}$ and $B = \{a, e, i, o, u\}$                                                                                      |
| 37    | If $U = \{a, b, c, d, e, f, g, h\}$ , $A = \{b, d, f, h\}$ and $B = \{a, d, e, h\}$ , find the following sets: (i) $A'$ (ii) $B'$ (iii) $A' \cup B'$ (iv) $A' \cap B'$                              |
| 38    | Find the symmetric difference between the following sets: $P = \{2, 3, 5, 7, 11\}$ and $Q = \{1, 3, 5, 11\}$                                                                                        |
| 39    | Find the symmetric difference between the following sets: $R = \{l, m, n, o, p\}$ and $S = \{j, l, n, q\}$                                                                                          |
| 40    | If $A = \{b, e, f, g\}$ and $B = \{c, e, g, h\}$ , then verify the commutative property of (i) union of sets (ii) intersection of sets.                                                             |
| 41    | Test for the commutative property of union and intersection of the sets $P = \{x : x \text{ is a real number between 2 and 7}\}$ and $Q = \{x : x \text{ is a rational number between 2 and 7}\}$ . |
| 42    | From the Venn diagram, verify that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ .                                                                                                                      |
| 43    | If $n(A) = 36, n(B) = 10, n(A \cup B) = 40$ and $n(A') = 27$ find $n(U)$ and $n(A \cap B)$ .                                                                                                        |
| 44    | Let $A = \{b, d, e, g, h\}$ and $B = \{a, e, c, h\}$ . Verify that $n(A - B) = n(A) - n(A \cap B)$ .                                                                                                |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

---

| S.No. | IMPORTANT QUESTIONS                                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 45    | If $n(A) = 25, n(B) = 40, n(A \cup B) = 50$ and $n(B') = 25$ , find $n(A \cap B)$ and $n(U)$ .                                                                   |
| 46    | If $n(A) = 300, n(A \cup B) = 500, n(A \cap B) = 50$ and $n(B') = 350$ find $n(B)$ and $n(U)$ .                                                                  |
| 47    | If $U = \{x : x \in \mathbb{N}, x \leq 10\}, A = \{2, 3, 4, 8, 10\}$ and $B = \{1, 2, 5, 8, 10\}$ , then verify that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ . |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics

#### Chapter 1: Set Language

#### 5 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | If $A = \{-\frac{1}{2}, 0, \frac{1}{4}, \frac{3}{4}, 2\}$ , $B = \{0, \frac{1}{4}, \frac{3}{4}, 2, \frac{5}{2}\}$ and $C = \{-\frac{1}{2}, \frac{1}{4}, 1, 2, \frac{5}{2}\}$ then verify $A \cap (B \cap C) = (A \cap B) \cap C$ . |
| 2     | If $P = \{1, 2, 5, 7, 9\}$ , $Q = \{2, 3, 5, 9, 11\}$ , $R = \{3, 4, 5, 7, 9\}$ and $S = \{2, 3, 4, 5, 8\}$ then find (i) $P \cup Q \cup R$ (ii) $P \cap Q \cap S$ (iii) $Q \cap S \cap R$ .                                       |
| 3     | If $A = \{p, q, r, s\}$ , $B = \{m, n, q, s, t\}$ and $C = \{m, n, p, q, s\}$ , then verify the associative property of union of sets.                                                                                             |
| 4     | Verify the associative property of intersection of sets for $A = \{-11, 2, 5, 7\}$ , $B = \{3, 5, 6, 13\}$ and $C = \{2, 3, 5, 9\}$ .                                                                                              |
| 5     | If $A = \{x : x = 2n, n \in \mathbb{W} \text{ and } n < 4\}$ , $B = \{x : x = 2n, n \in \mathbb{N} \text{ and } n \leq 4\}$ and $C = \{0, 1, 2, 5, 6\}$ , then verify the associative property of intersection of sets.            |
| 6     | If $A = \{0, 2, 4, 6, 8\}$ , $B = \{x : x \text{ is a prime number and } x < 11\}$ and $C = \{x : x \in \mathbb{N} \text{ and } 5 \leq x < 9\}$ then verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ .                     |
| 7     | Verify $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using Venn diagrams.                                                                                                                                                       |
| 8     | Verify $A - (B \cup C) = (A - B) \cap (A - C)$ using Venn diagrams.                                                                                                                                                                |
| 9     | If $P = \{x : x \in \mathbb{N} \text{ and } 0 < x < 10\}$ , $Q = \{x : x = 2n + 1, n \in \mathbb{W} \text{ and } n < 5\}$ and $R = \{2, 3, 5, 7, 11, 13\}$ , then verify $P - (Q \cap R) = (P - Q) \cap (P - R)$ .                 |
| 10    | Verify $(A \cup B)' = A' \cap B'$ using Venn diagrams.                                                                                                                                                                             |
| 11    | If $U = \{x : x \in \mathbb{Z}, -2 \leq x \leq 10\}$ , $A = \{x : x = 2p + 1, p \in \mathbb{Z}, -1 \leq p \leq 4\}$ , $B = \{x : x = 3q + 1, q \in \mathbb{Z}, -1 \leq q < 4\}$ , verify De Morgan's laws for complementation.     |
| 12    | If $K = \{a, b, d, e, f\}$ , $L = \{b, c, d, g\}$ and $M = \{a, b, c, d, h\}$ , then find (i) $K \cup (L \cap M)$ (ii) $(K \cap L) \cup M$ and verify distributive laws.                                                           |
| 13    | If $A = \{x : x \in \mathbb{Z}, -2 < x \leq 4\}$ , $B = \{x : x \in \mathbb{W}, x \leq 5\}$ , $C = \{-4, -1, 0, 2, 3, 4\}$ , then verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ .                                        |
| 14    | Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using Venn diagrams.                                                                                                                                                       |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | If $A = \{b, c, e, g, h\}$ , $B = \{a, c, d, g, i\}$ and $C = \{a, d, e, g, h\}$ , then show that $A - (B \cap C) = (A - B) \cup (A - C)$ .                                                                                                                                                                                                                                                     |
| 16    | If $A = \{x : x = 6n, n \in \mathbb{N} \text{ and } n < 6\}$ , $B = \{x : x = 2n, n \in \mathbb{N} \text{ and } 2 < n \leq 9\}$ and $C = \{x : x = 3n, n \in \mathbb{N} \text{ and } 4 \leq n < 10\}$ , then show that $A - (B \cap C) = (A - B) \cup (A - C)$ .                                                                                                                                |
| 17    | If $A = \{-2, 0, 1, 3, 5\}$ , $B = \{-1, 0, 2, 5, 6\}$ and $C = \{-1, 2, 5, 6, 7\}$ , then show that $A - (B \cup C) = (A - B) \cap (A - C)$ .                                                                                                                                                                                                                                                  |
| 18    | If $A = \{y : y = \frac{a+1}{2}, a \in \mathbb{W} \text{ and } a \leq 5\}$ , $B = \{y : y = \frac{2n-1}{2}, n \in \mathbb{N} \text{ and } n < 5\}$ and $C = \{-1, -\frac{1}{2}, 1, \frac{3}{2}, 2\}$ , then show that $A - (B \cup C) = (A - B) \cap (A - C)$ .                                                                                                                                 |
| 19    | Verify $A - (B \cap C) = (A - B) \cup (A - C)$ using Venn diagrams.                                                                                                                                                                                                                                                                                                                             |
| 20    | If $U = \{4, 7, 8, 10, 11, 12, 15, 16\}$ , $A = \{7, 8, 11, 12\}$ and $B = \{4, 8, 12, 15\}$ then verify De Morgan's Laws for complementation.                                                                                                                                                                                                                                                  |
| 21    | Verify $(A \cap B)' = A' \cup B'$ using Venn diagrams.                                                                                                                                                                                                                                                                                                                                          |
| 22    | In a school, all students play either Hockey or Cricket or both. 300 play Hockey, 250 play Cricket and 110 play both games. Find<br>(i) the number of students who play only Hockey<br>(ii) the number of students who play only Cricket<br>(iii) the total number of students in the School.                                                                                                   |
| 23    | In a party of 60 people, 35 had Vanilla ice cream, 30 had Chocolate ice cream. All the people had at least one ice cream. Then how many of them had<br>(i) both Vanilla and Chocolate ice cream<br>(ii) only Vanilla ice cream<br>(iii) only Chocolate ice cream.                                                                                                                               |
| 24    | In a college, 240 students play cricket, 180 students play football, 164 students play hockey, 42 play both cricket and football, 38 play both football and hockey, 40 play both cricket and hockey and 16 play all the three games. If each student participate in atleast one game, then find (i) the number of students in the college (ii) the number of students who play only one game.   |
| 25    | In a residential area with 600 families 35 owned scooter, 13 owned car, 14 owned bicycle, 120 families owned scooter and car, 86 owned car and bicycle while 90 families owned scooter and bicycle. If $\frac{2}{15}$ of families owned all the three types of vehicles, then find (i) the number of families owned atleast two types of vehicle. (ii) the number of families owned no vehicle. |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26    | In a group of 100 students, 85 students speak Tamil, 40 students speak English, 20 students speak French, 32 speak Tamil and English, 13 speak English and French and 10 speak Tamil and French. If each student knows atleast any one of these languages, then find the number of students who speak all these three languages.                                                                                      |
| 27    | A survey was conducted among 200 magazine subscribers of three different magazines A, B and C. It was found that 75 members do not subscribe magazine A, 100 members do not subscribe magazine B, 50 members do not subscribe magazine C and 125 subscribe atleast two of the three magazines. Find (i) Number of members who subscribe exactly two magazines (ii) Number of members who subscribe only one magazine. |
| 28    | Verify $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$ for the following sets: $A = \{a, c, e, f, h\}$ , $B = \{c, d, e, f\}$ and $C = \{a, b, c, f\}$ .                                                                                                                                                                                                     |
| 29    | Verify $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$ for $A = \{1, 3, 5\}$ , $B = \{2, 3, 5, 6\}$ and $C = \{1, 5, 6, 7\}$ .                                                                                                                                                                                                                               |
| 30    | In a class, all students take part in either music or drama or both. 25 students take part in music, 30 students take part in drama and 8 students take part in both music and drama. Find (i) The number of students who take part in only music (ii) The number of students who take part in only drama (iii) The total number of students in the class.                                                            |
| 31    | In a party of 45 people, each one likes tea or coffee or both. 35 people like tea and 20 people like coffee. Find the number of people who (i) like both tea and coffee (ii) do not like tea (iii) do not like coffee.                                                                                                                                                                                                |
| 32    | In an examination 50% of the students passed in Mathematics and 70% of students passed in Science while 10% students failed in both subjects. 300 students passed in both the subjects. Find the total number of students who appeared in the examination, if they took examination in only two subjects.                                                                                                             |
| 33    | A and B are two sets such that $n(A - B) = 32 + x$ , $n(B - A) = 5x$ and $n(A \cap B) = x$ . Illustrate the information by means of a Venn diagram. Given that $n(A) = n(B)$ , calculate the value of $x$ .                                                                                                                                                                                                           |
| 34    | Out of 500 car owners investigated, 400 owned car A and 200 owned car B, 50 owned both A and B cars. Is this data correct?                                                                                                                                                                                                                                                                                            |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 35    | In a colony, 275 families buy Tamil newspaper, 150 families buy English newspaper, 45 families buy Hindi newspaper, 125 families buy Tamil and English newspapers, 17 families buy English and Hindi newspapers, 5 families buy Tamil and Hindi newspapers and 3 families buy all the three newspapers. If each family buy atleast one of these newspapers then find (i) Number of families buy only one newspaper (ii) Number of families buy atleast two newspapers (iii) Total number of families in the colony. |
| 36    | A survey of 1000 farmers found that 600 grew paddy, 350 grew ragi, 280 grew corn, 120 grew paddy and ragi, 100 grew ragi and corn, 80 grew paddy and corn. If each farmer grew atleast any one of the above three, then find the number of farmers who grew all the three.                                                                                                                                                                                                                                          |
| 37    | Each student in a class of 35 plays atleast one game among chess, carrom and table tennis. 22 play chess, 21 play carrom, 15 play table tennis, 10 play chess and table tennis, 8 play carrom and table tennis and 6 play all the three games. Find the number of students who play (i) chess and carrom but not table tennis (ii) only chess (iii) only carrom.                                                                                                                                                    |
| 38    | In a class of 50 students, each one come to school by bus or by bicycle or on foot. 25 by bus, 20 by bicycle, 30 on foot and 10 students by all the three. Now how many students come to school exactly by two modes of transport?                                                                                                                                                                                                                                                                                  |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics

#### Chapter 2: Real Numbers

#### 2 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Find any two rational numbers between $\frac{1}{2}$ and $\frac{2}{3}$ .                                                                                                                            |
| 2     | Find any three rational numbers between $\frac{-7}{11}$ and $\frac{2}{11}$ .                                                                                                                       |
| 3     | Find any five rational numbers between: (i) $\frac{1}{4}$ and $\frac{1}{5}$ (ii) 0.1 and 0.11 (iii) $-1$ and $-2$ .                                                                                |
| 4     | Represent the following as decimal form: (i) $\frac{-4}{11}$ (ii) $\frac{11}{75}$ .                                                                                                                |
| 5     | Convert the following decimal numbers in the form of $\frac{p}{q}$ : (i) 0.35 (ii) 2.176 (iii) $-0.0028$ .                                                                                         |
| 6     | Without actual division, classify the decimal expansion of the following numbers as terminating or non-terminating and recurring: (i) $\frac{13}{64}$ (ii) $\frac{-71}{125}$ .                     |
| 7     | Verify that $1 = 0.\overline{9}$ .                                                                                                                                                                 |
| 8     | Express $\frac{1}{13}$ in decimal form. Find the length of the period of decimals.                                                                                                                 |
| 9     | Express the following decimal expression into rational numbers: (i) $0.\overline{24}$ (ii) $3.\overline{17}$ .                                                                                     |
| 10    | Without actual division, find which of the following rational numbers have terminating decimal expansion: (i) $\frac{7}{128}$ (ii) $\frac{21}{15}$ (iii) $4\frac{9}{35}$ (iv) $\frac{219}{2200}$ . |
| 11    | Classify the numbers as rational or irrational: (i) $\sqrt{10}$ (ii) $\sqrt{49}$ (iii) 0.025 (iv) $0.\overline{76}$ .                                                                              |
| 12    | Find any 3 irrational numbers between 0.12 and 0.13.                                                                                                                                               |
| 13    | Give any two rational numbers lying between $0.5151151115 \dots$ and $0.5353353335 \dots$                                                                                                          |
| 14    | Find any two irrational numbers between $\frac{6}{7}$ and $\frac{12}{13}$ .                                                                                                                        |
| 15    | Find any two rational numbers between $2.2360679 \dots$ and $2.236505500 \dots$                                                                                                                    |
| 16    | Express the following in the form $2^n$ : (i) 8 (ii) 32 (iii) $\frac{1}{4}$ (iv) $\sqrt{2}$ (v) $\sqrt{8}$ .                                                                                       |
| 17    | Find the value of: (i) $(81)^{\frac{5}{4}}$ (ii) $(64)^{\frac{-2}{3}}$ .                                                                                                                           |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18    | Write the following in the form of $5^n$ : (i) 625 (ii) $\frac{1}{5}$ (iii) $\sqrt{5}$ (iv) $\sqrt{125}$ .                                                                |
| 19    | Write the following in the form of $4^n$ : (i) 16 (ii) 8 (iii) 32.                                                                                                        |
| 20    | Find the value of: (i) $(49)^{\frac{1}{2}}$ (ii) $(243)^{\frac{2}{5}}$ (iii) $9^{-\frac{3}{2}}$ (iv) $(\frac{64}{125})^{-\frac{2}{3}}$ .                                  |
| 21    | Use a fractional index to write: (i) $\sqrt{5}$ (ii) $\sqrt[2]{7}$ (iii) $(\sqrt[3]{49})^5$ (iv) $(\sqrt[3]{\frac{1}{100}})^7$ .                                          |
| 22    | Find the 5th root of: (i) 32 (ii) 243 (iii) 100000 (iv) $\frac{1024}{3125}$ .                                                                                             |
| 23    | Reduce the following numbers to surds of same order: (i) $\sqrt{3}$ (ii) $\sqrt[4]{3}$ (iii) $\sqrt[3]{3}$ .                                                              |
| 24    | Express the surds in the simplest form: (i) $\sqrt{8}$ (ii) $\sqrt[3]{192}$ .                                                                                             |
| 25    | Show that $\sqrt[3]{7} > \sqrt[4]{5}$ .                                                                                                                                   |
| 26    | Express each of the following surds in its simplest form and find its order, radicand and coefficient: (i) $\sqrt[3]{108}$ (ii) $\sqrt[3]{(1024)^{-2}}$ .                 |
| 27    | (i) Add $3\sqrt{7}$ and $5\sqrt{7}$ . Check whether the sum is rational or irrational. (ii) Subtract $4\sqrt{5}$ from $7\sqrt{5}$ . Is the answer rational or irrational? |
| 28    | Multiply $\sqrt[3]{40}$ and $\sqrt[3]{16}$ .                                                                                                                              |
| 29    | Divide $\sqrt[3]{98}$ by $\sqrt[3]{66}$ .                                                                                                                                 |
| 30    | Rationalise the denominator of: (i) $\frac{7}{\sqrt{14}}$ (ii) $\frac{5+\sqrt{3}}{5-\sqrt{3}}$ .                                                                          |
| 31    | Rationalise the denominator: (i) $\frac{1}{\sqrt{50}}$ (ii) $\frac{5}{\sqrt[3]{5}}$ (iii) $\frac{\sqrt{75}}{\sqrt{18}}$ (iv) $\frac{3\sqrt{5}}{\sqrt{6}}$ .               |
| 32    | Express in scientific notation: (i) 9768854 (ii) 0.04567891 (iii) 72006865.48.                                                                                            |
| 33    | Write the following numbers in decimal form: (i) $6.34 \times 10^4$ (ii) $2.00367 \times 10^{-5}$ .                                                                       |
| 34    | The mass of the Earth is $5.97 \times 10^{24}$ kg and that of the Moon is $0.073 \times 10^{24}$ kg. What is their total mass?                                            |
| 35    | Write the following in scientific notation: (i) 500000004 (ii) 0.000000053 (iii) $300000^3 \times 2000^4$ .                                                               |
| 36    | Represent the following numbers in the scientific notation: (i) 569430000000 (ii) 2000.57 (iii) 0.0000006000 (iv) 0.0009000002.                                           |
| 37    | Write the following numbers in decimal form: (i) $3.459 \times 10^6$ (ii) $5.678 \times 10^4$ (iii) $1.00005 \times 10^{-5}$ (iv) $2.530009 \times 10^{-7}$ .             |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

---

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38    | Represent the following information in scientific notation: (i) The world population is nearly 7000,000,000. (ii) One light year means the distance 9460528400000000 km. |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics

#### Chapter 2: Real Numbers

#### 5 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Express the rational number $\frac{1}{27}$ in recurring decimal form by using the recurring decimal expansion of $\frac{1}{3}$ . Hence write $\frac{59}{27}$ in recurring decimal form.                        |
| 2     | Convert the following decimal numbers in the form of $\frac{p}{q}$ ( $p, q \in \mathbb{Z}$ and $q \neq 0$ ):<br>(i) $0.\overline{3}$ (ii) $2.\overline{124}$ (iii) $0.4\overline{5}$ (iv) $0.5\overline{68}$ . |
| 3     | Express the following rational numbers into decimal and state the kind of decimal expansion: (i) $\frac{2}{7}$ (ii) $-5\frac{3}{11}$ (iii) $\frac{22}{3}$ (iv) $\frac{327}{200}$ .                             |
| 4     | Express the rational number $\frac{1}{33}$ in recurring decimal form by using the recurring decimal expansion of $\frac{1}{11}$ . Hence write $\frac{71}{33}$ in recurring decimal form.                       |
| 5     | Express the following decimal expression into rational numbers: (i) $2.3\overline{27}$ (ii) $-5.13\overline{2}$ (iii) $17.21\overline{5}$ (iv) $-21.21\overline{37}$ .                                         |
| 6     | Find the decimal expansion of $\sqrt{3}$ .                                                                                                                                                                     |
| 7     | Find whether $x$ and $y$ are rational or irrational in the following: (i) $a = 2 + \sqrt{3}, b = 2 - \sqrt{3}; x = a + b, y = a - b$ (ii) $a = \sqrt{2} + 7, b = \sqrt{2} - 7; x = a + b, y = a - b$ .         |
| 8     | Find whether $x$ and $y$ are rational or irrational in the following: (i) $a = \sqrt{75}, b = \sqrt{3}; x = ab, y = \frac{a}{b}$ (ii) $a = \sqrt{18}, b = \sqrt{3}; x = ab, y = \frac{a}{b}$ .                 |
| 9     | Represent $\sqrt{9.3}$ on a number line.                                                                                                                                                                       |
| 10    | Represent the following irrational numbers on the number line: (i) $\sqrt{3}$ (ii) $\sqrt{4.7}$ (iii) $\sqrt{6.5}$ .                                                                                           |
| 11    | Represent 4.863 on the number line.                                                                                                                                                                            |
| 12    | Represent 3.45 on the number line upto 4 decimal places.                                                                                                                                                       |
| 13    | Represent the following numbers on the number line: (i) 5.348 (ii) $6.\overline{4}$ upto 3 decimal places (iii) $4.\overline{73}$ upto 4 decimal places.                                                       |
| 14    | Arrange in ascending order: $\sqrt[3]{2}, \sqrt[2]{4}, \sqrt[4]{3}$ .                                                                                                                                          |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

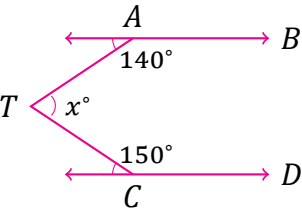
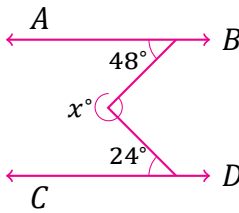
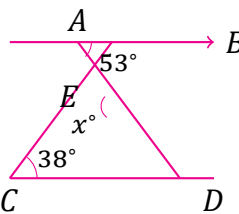
| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | Simplify the following: (i) $\sqrt{63} - \sqrt{175} + \sqrt{28}$ (ii) $2\sqrt[3]{40} + 3\sqrt[3]{625} - 4\sqrt[3]{320}$ .                                                                                                                                                        |
| 16    | Compute and give the answer in the simplest form: $2\sqrt{72} \times 5\sqrt{32} \times 3\sqrt{50}$ .                                                                                                                                                                             |
| 17    | Simplify the following using addition and subtraction properties of surds: (i) $5\sqrt{3} + 18\sqrt{3} - 2\sqrt{3}$ (ii) $4\sqrt[3]{5} + 2\sqrt[3]{5} - 3\sqrt[3]{5}$ (iii) $3\sqrt{75} + 5\sqrt{48} - 2\sqrt{243}$ .                                                            |
| 18    | Simplify the following using multiplication and division properties of surds: (i) $\sqrt{3} \times \sqrt{5} \times \sqrt{2}$ (ii) $\sqrt{35} \div \sqrt{7}$ (iii) $\sqrt[3]{27} \times \sqrt[3]{8} \times \sqrt[3]{125}$ (iv) $(7\sqrt{a} - 5\sqrt{b})(7\sqrt{a} + 5\sqrt{b})$ . |
| 19    | If $\sqrt{2} = 1.414, \sqrt{3} = 1.732, \sqrt{5} = 2.236, \sqrt{10} = 3.162$ , then find the values of the following correct to 3 places of decimals: (i) $\sqrt{40} - \sqrt{20}$ (ii) $\sqrt{300} + \sqrt{90} - \sqrt{8}$ .                                                     |
| 20    | Arrange surds in descending order: (i) $\sqrt[3]{5}, \sqrt[9]{4}, \sqrt[6]{3}$ (ii) $\sqrt[2]{\sqrt[3]{5}}, \sqrt[3]{\sqrt[4]{7}}, \sqrt{\sqrt{3}}$ .                                                                                                                            |
| 21    | Rationalise the denominator and simplify: (i) $\frac{\sqrt{48} + \sqrt{32}}{\sqrt{27} - \sqrt{18}}$ (ii) $\frac{5\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$ .                                                                                                                    |
| 22    | Rationalise the denominator and simplify: (i) $\frac{2\sqrt{6} - \sqrt{5}}{3\sqrt{5} - 2\sqrt{6}}$ (ii) $\frac{\sqrt{5}}{\sqrt{6} + \sqrt{2}} - \frac{\sqrt{5}}{\sqrt{6} - \sqrt{2}}$ .                                                                                          |
| 23    | Find the value of $a$ and $b$ if $\frac{\sqrt{7}-2}{\sqrt{7}+2} = a\sqrt{7} + b$ .                                                                                                                                                                                               |
| 24    | If $x = \sqrt{5} + 2$ , then find the value of $x^2 + \frac{1}{x^2}$ .                                                                                                                                                                                                           |
| 25    | Given $\sqrt{2} = 1.414$ , find the value of $\frac{8-5\sqrt{2}}{3-2\sqrt{2}}$ (to 3 places of decimals).                                                                                                                                                                        |
| 26    | Represent the following numbers in scientific notation: (i) $(300000^2 \times 20000^4)$ (ii) $0.00000111 \div 0.0053$ .                                                                                                                                                          |
| 27    | Represent the following numbers in scientific notation: $\frac{(0.000036 \times 0.000054)}{(0.0093 \times 0.052)}$ .                                                                                                                                                             |
| 28    | Simplify: (i) $2.75 \times 10^7 + 1.23 \times 10^8$ (ii) $1.598 \times 10^{17} - 4.58 \times 10^{15}$ .                                                                                                                                                                          |
| 29    | Simplify: (i) $1.02 \times 10^{10} \times 1.20 \times 10^{-3}$ (ii) $8.41 \times 10^4 \div 4.3 \times 10^5$ .                                                                                                                                                                    |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics Chapter 3 & 4: Algebra and Geometry Important 2 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------|
| 1     | Find the degree of the polynomial: $6ab^8 + 5a^2b^3c^2 - 7ab + 4b^2c + 2$ .                               |
| 2     | Rewrite the following polynomial in standard form: $x - 9 + 7x^3 + 6x^2$ .                                |
| 3     | Add the following polynomials: $p(x) = 6x^2 - 7x + 2$ and $q(x) = 6x^3 - 7x + 15$ .                       |
| 4     | Subtract the polynomial $q(x) = 6x - 9$ from $p(x) = 7x^2 + 6x - 1$ .                                     |
| 5     | Find the product of $(4x - 5)$ and $(2x^2 + 3x - 6)$ .                                                    |
| 6     | Find the value of the polynomial $f(y) = 6y - 3y^2 + 3$ at $y = -1$ .                                     |
| 7     | Find the zeros of the polynomial: $p(x) = 2x + 5$ .                                                       |
| 8     | Find the roots of the polynomial equation: $5x - 6 = 0$ .                                                 |
| 9     | Check whether $-3$ and $3$ are zeros of the polynomial $x^2 - 9$ .                                        |
| 10    | Find the remainder when $f(x) = x^3 + 3x^2 + 3x + 1$ is divided by $x + 1$ .                              |
| 11    | By remainder theorem, find the remainder when $p(x) = x^3 - 2x^2 - 4x - 1$ is divided by $g(x) = x + 1$ . |
| 12    | Determine whether $x - 1$ is a factor of the polynomial: $x^3 + 5x^2 - 10x + 4$ .                         |
| 13    | Find the value of $m$ , if $x - 2$ is a factor of the polynomial $2x^3 - 6x^2 + mx + 4$ .                 |
| 14    | Expand the following using identities: $(3x + 4y)^2$ .                                                    |
| 15    | Expand: $(a - b + c)^2$ .                                                                                 |
| 16    | Find the area of a square whose side length is $3m + 2n - 4l$ .                                           |
| 17    | Evaluate the following by using identities: $98^3$ .                                                      |
| 18    | Factorise the following: $9x^2 + 12xy + 4y^2$ .                                                           |
| 19    | Factorise the following: $x^2 + 4x + 4$ .                                                                 |
| 20    | Factorise the following: $8x^3 + 125y^3$ .                                                                |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

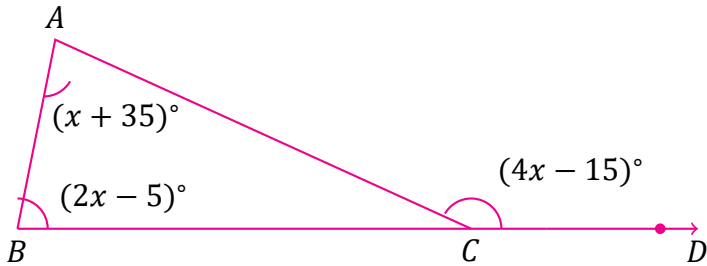
| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21    | Divide $x^3 - 4x^2 + 6x$ by $x$ , where $x \neq 0$ .                                                                                                                                |
| 22    | Find the quotient and the remainder when $(5x^2 - 7x + 2) \div (x - 1)$ .                                                                                                           |
| 23    | Find GCD of the following: $16x^3y^2, 24xy^3z$ .                                                                                                                                    |
| 24    | Find the GCD for the following: $p^5, p^{11}, p^9$ .                                                                                                                                |
| 25    | Find the GCD of the following: $35x^5y^3z^4, 49x^2yz^3, 14xy^2z^2$ .                                                                                                                |
| 26    | Find the GCD of the following: $x^4 - 1, x^2 - 1$ .                                                                                                                                 |
| 27    | The angles of a triangle are in the ratio 1: 2: 3, find the measure of each angle of the triangle.                                                                                  |
| 28    | <p>In the figure, <math>AB</math> is parallel to <math>CD</math>, find <math>x^\circ</math></p>   |
| 29    | <p>In the figure, <math>AB</math> is parallel to <math>CD</math>, find <math>x^\circ</math></p>  |
| 30    | <p>In the figure, <math>AB</math> is parallel to <math>CD</math>, find <math>x^\circ</math></p>  |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

### 9th Standard Mathematics Chapter 3 & 4: Algebra and Geometry Important 5 Mark Question Bank

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | What should be added to $2x^3 + 6x^2 - 5x + 8$ to get $3x^3 - 2x^2 + 6x + 15$ ?                                                                                           |
| 2     | What must be subtracted from $2x^4 + 4x^2 - 3x + 7$ to get $3x^3 - x^2 + 2x + 1$ ?                                                                                        |
| 3     | For what value of $k$ is the polynomial $2x^4 + 3x^3 + 2kx^2 + 3x + 6$ exactly divisible by $x + 2$ ?                                                                     |
| 4     | If two polynomials $2x^3 + ax^2 + 4x - 12$ and $x^3 + x^2 - 2x + a$ leave the same remainder when divided by $x - 3$ , find the value of $a$ and also find the remainder. |
| 5     | Using factor theorem, show that $x - 5$ is a factor of the polynomial $2x^3 - 5x^2 - 28x + 15$ .                                                                          |
| 6     | Determine the value of $m$ , if $x + 3$ is a factor of $x^3 - 3x^2 - mx + 24$ .                                                                                           |
| 7     | Check if $x + 2$ and $x - 4$ are the sides of a rectangle whose area is $x^2 - 2x - 8$ by using factor theorem.                                                           |
| 8     | Expand: $(2x + 3y + 4z)^2$ .                                                                                                                                              |
| 9     | Find the product of $(2x + 3y + 4z)(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8zx)$ .                                                                                            |
| 10    | If $x + y + z = 9$ and $xy + yz + zx = 26$ , then find the value of $x^2 + y^2 + z^2$ .                                                                                   |
| 11    | Find $27a^3 + 64b^3$ if $3a + 4b = 10$ and $ab = 2$ .                                                                                                                     |
| 12    | Factorise the following: $8x^3 + 27y^3 + 64z^3 - 72xyz$ .                                                                                                                 |
| 13    | Factorise: $2x^2 + 15x + 27$ .                                                                                                                                            |
| 14    | Factorise: $(x + y)^2 + 9(x + y) + 20$ .                                                                                                                                  |
| 15    | Factorise the following: $5x^2 - 29xy - 42y^2$ .                                                                                                                          |
| 16    | Find the quotient and remainder when $3x^3 - 4x^2 - 5$ is divided by $3x + 1$ using synthetic division.                                                                   |
| 17    | If the quotient obtained on dividing $8x^4 - 2x^2 + 6x - 7$ by $2x + 1$ is $4x^3 + px^2 - qx + 3$ , then find $p, q$ and also the remainder.                              |

## 9 TH MATHS QUARTERLY QUESTION BANK 2026

| S.No. | IMPORTANT QUESTIONS                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18    | Factorise the following polynomial using synthetic division: $x^3 - 3x^2 - 10x + 24$ .                                                                                                                              |
| 19    | Factorise each of the following polynomials using synthetic division: $x^3 - 7x + 6$ .                                                                                                                              |
| 20    | Find the GCD of the following: $a^3 - 9ax^2, (a - 3x)^2$ .                                                                                                                                                          |
| 21    | $\triangle ABC$ and $\triangle DEF$ are two triangles in which $AB = DF, \angle ACB = 70^\circ, \angle ABC = 60^\circ; \angle DEF = 70^\circ$ and $\angle EDF = 60^\circ$ . Prove that the triangles are congruent. |
| 22    | Find all the three angles of the $\triangle ABC$<br>                                                                              |